

# Project Instructions — Stage 04: Data Acquisition and Ingestion

---

All paths below are inside `project/`.

Today's Project Contribution:

Today you'll complete a piece of your full data project.

This task aligns with the **Data Acquisition/Ingestion** lifecycle stage, where you will:

- Acquire data programmatically using an API and/or scraping a permitted public table.
- Add raw data and ingestion scripts/notebooks to your existing project repo.

By the end of this assignment, your project should include the elements listed below.

**Deliverable Options (from Stage Scaffold):**

**Required:**

- `.env` file storing API keys, tokens, or file paths (**not committed**); include `.env.example` in the repo.
- Raw data saved in `data/raw/` folder (CSV, JSON, etc.).
- `notebooks/project_pipeline.ipynb` — your project's pipeline notebook, started this stage and extended in every stage from here on. Add the steps that acquire your data and save it into `data/raw/`, then run the whole notebook top to bottom before you commit.

Start `project_pipeline.ipynb` with this cell, so it works no matter where it is run from:

```
# --- run me first ---
from pathlib import Path
import os, sys
if Path.cwd().name == 'notebooks':
    os.chdir('..')           # project/notebooks -> project
ROOT = Path.cwd()
if str(ROOT) not in sys.path:
    sys.path.insert(0, str(ROOT)) # so `from src....` imports work
print('working from:', ROOT.name)
```

**Optional (Choose One):**

- API pull script.
- Web scraping notebook.
- Pre-collected dataset with short written justification.

**How This Fits Into Your Final Project:**

Your work today builds toward a complete, end-to-end project by establishing a **trusted raw data foundation** for later EDA and modeling.

**Before next class:**

- Save files to appropriate folders (`data/raw/`, `src/`, `notebooks/`).

- Commit and push changes to GitHub. Ensure `.env` is ignored.
- Document sources, parameters, and validations in your README or notebook.

**Explicit Chain:**

In your **homework**, you produced an **API pull and a scraped table with validations**.

**Now, you will adapt that work to formalize your project's ingestion layer**, ensuring raw files are saved with reproducible filenames, secrets are managed via `.env`, and documentation is updated.

**By the end of the course**, your full project will follow the complete lifecycle.